

Chapter 4: Requirements Engineering

Requirements & Types

Requirements

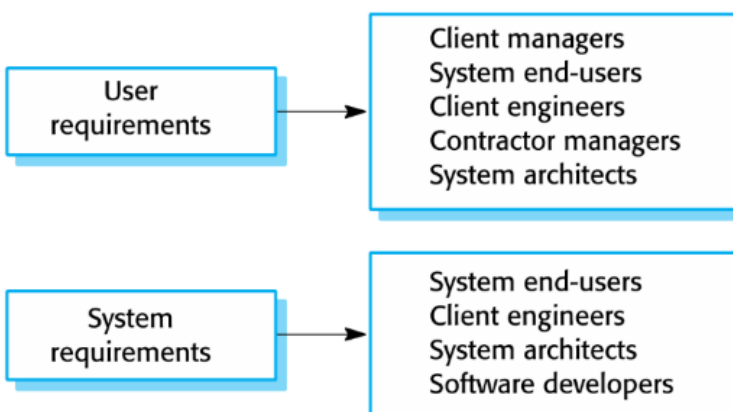
- **Services a customer requires** from a system and the **constraints** under which it operates.
- Can range from high-level statements to detailed specifications.
- Describe **what the system should do**, not how it should do it.

User Requirements

- Written in **natural language** and **diagrams**.
- **High-level description** for customers and managers.
- Easy for non-technical users to understand.

System Requirements

- **Detailed** and **structured description** of functions, services, and constraints.
- Defines **exactly what should be implemented**.
- May be **part** of the **development contract**.



Stakeholders: Anyone affected by the system.

- End users
 - Managers
 - Owners
 - External stakeholders (regulators, maintenance staff, etc.)
-

Functional Requirements (FR)

- Describe **services** and **behavior** of the system.
- **what system should do.**
- May also specify what system should not do.
- Examples: Login, Search records, Generate reports, Register users

Problems with Functional Requirements

Imprecision (Ambiguity)

- Different people may **interpret requirements differently.**

Completeness

- **All required functions** should be **included.**

Consistency

- Requirements **should not conflict** with each other.
-

Non-Functional Requirements (NFR)

Constraints on system operation or development.

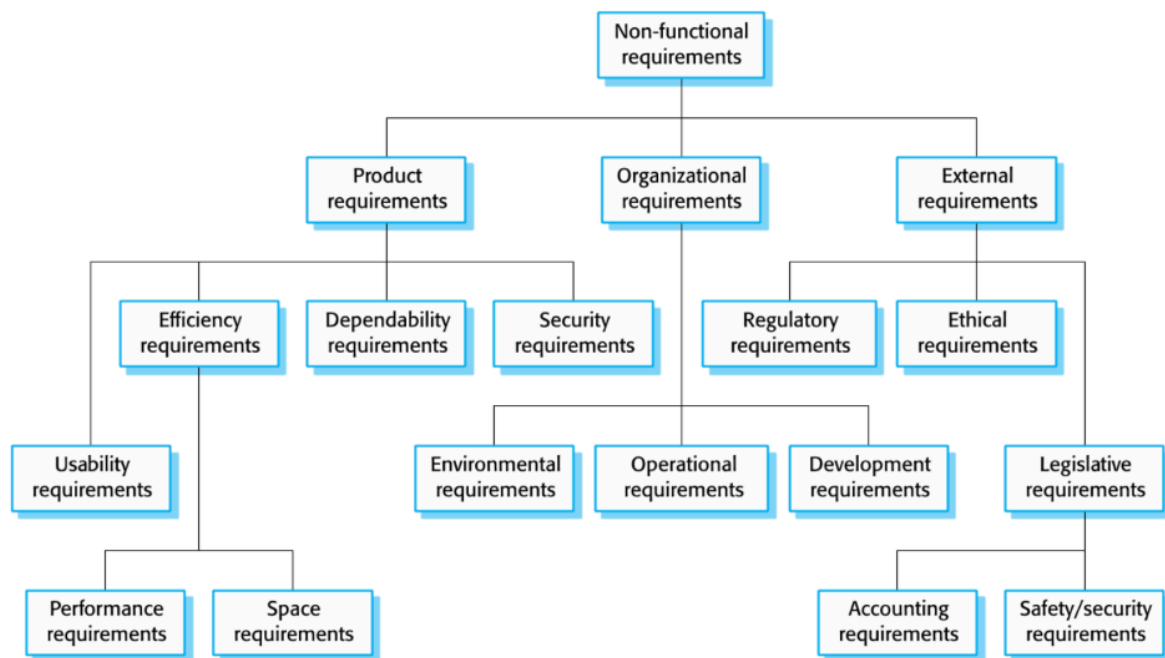
- Performance

- Security
- Reliability
- Availability
- Standards

Important:

- Often **apply** to the **whole system**.
- **Sometimes** more critical than functional requirements.

NFR Type	Description	Example
Product	Runtime behavior constraints > specify that the delivered product must behave in a particular way	System available Mon–Fri, downtime ≤ 5 sec/day
Organizational	Derived from company policies	Login using ID card
External	Derived from laws/regulations	Follow patient privacy laws



Goals vs Verifiable Requirements

Goal

- General intention.

Verifiable Requirement

- Can be measured and tested.

NFR Metrics

Attribute	Measure
Speed	Response time, transactions/sec
Reliability	Failure rate, availability
Ease of Use	Training time
Robustness	Restart time after failure
Portability	Number of supported systems
size	No of bytes

Requirements Engineering Process

RE is **iterative** (activities repeat throughout development).

Main Activities

1. **Feasibility Study**
2. **Requirements Elicitation**
3. **Requirements Analysis**

4. Requirements Specification
 5. Requirements Validation
 6. Requirements Management
-

Requirements Elicitation & Analysis / discovery

Process of **discovering system requirements** by interacting with stakeholders.

Stages

1. Requirements Discovery

- Gather requirements from stakeholders.

2. Classification & Organization

- Group related requirements together.

3. Prioritization & Negotiation

- Resolve conflicts.
- Rank requirements by importance.

4. Requirements Specification

- Document the requirements.
-

Requirements Discovery Techniques

Interviewing

Closed Interview

- Predetermined questions.

Open Interview

- Free discussion.

Good practices:

- Stay open-minded.
- Listen carefully.
- Use examples or prototypes.

Problems:

- Stakeholders may not know what they want.
 - Domain terminology may be difficult.
 - Some knowledge is never explicitly stated.
-

Ethnography

- **Observe people while they work.**
- Reveals real work practices.
- Finds hidden requirements.

Advantages:

- Good for understanding existing processes.

Limitation:

- Cannot suggest completely new features.
-

Scenarios / User Stories

Real-life examples showing system usage.

A scenario includes:

- Starting situation
- Normal flow
- What can go wrong
- Concurrent activities
- Final state

Benefits:

- Easy for stakeholders to understand.
 - Encourages feedback and discussion.
-

Common Problems in Elicitation

- Stakeholders don't know exactly what they need.
 - Different stakeholders have **conflicting** requirements.
 - Political/organizational issues influence requirements.
 - Requirements change over time.
 - New stakeholders may appear later.
-

Requirements Specification

Process of **documenting requirements** in a formal requirements document.

Specification Methods

Natural Language

- Simple sentences.
- Most common method.

Structured Natural Language

- Uses templates/forms.

Graphical Models

- UML diagrams.
- Use cases.

Mathematical Specifications

- Formal mathematical notation.
- Very precise but difficult for customers.

Guidelines for Writing Requirements

- Use **standard format**.
- Use **consistent wording**.
- Use **shall** for **mandatory** requirements.
- Use **should** for **desirable** requirements.
- Avoid jargon.
- Explain **why requirement exists**.

Problems with Natural Language

- **Lack of clarity**
- Requirement **confusion**
- Requirement **amalgamation** (**multiple requirements combined together**)

Use Cases

- Describe **interactions between users (actors) and system**.

- Show **system functionality** from **user's perspective**.
 - Usually represented using **UML** diagrams.
-

Software Requirements Specification (SRS)

Official document describing system requirements.

Contains:

- **User requirements**
- **System requirements**

Purpose:

- **Agreement** between **customer** and **developer**.
 - Focuses on **WHAT** the system should do
-

Requirements Validation

Checking whether **requirements describe the system the customer actually wants**.

- Fixing requirement errors after delivery can cost up to **100× more** than fixing implementation errors.

Validation Checks

- **Validity** → **Right requirements?**
- **Consistency** → **No conflicts?**
- **Completeness** → **Nothing missing?**

- **Realism** → Can be built?
- **Verifiability** → Can be tested?

Validation Techniques

- **Requirements Reviews**
 - **Prototyping**
 - **Test Case Generation / TDD**
-

Requirements Change Management

Why Requirements Change

- Business environment changes.
- Technology changes.
- New regulations/laws appear.
- Customers and users have different needs.
- Large systems have conflicting user requirements.

Final requirements are usually a compromise.

Requirements Management

Process of **controlling and tracking requirement changes during development.**

Requirements Management Planning

Requirements identification (unique IDs), change management process (assess impact & cost), traceability policies, tool support.

Requirements Identification

- Each requirement gets a unique ID.

Change Management Process

- Assess **impact** and **cost** of **changes**.

Traceability

- **Track relationships** between **requirements** and **design**.

Tool Support

- **Databases, spreadsheets, requirement management tools.**
-